

Model-Driven Neural Network Based MIMO Channel Estimator

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Abstract

Index Terms

I. INTRODUCTION

II. METHODOLOGY

A. Uplink Latency Minimization

For MIMO uplink (UL), the receive signal is written as

$$\mathbf{y} = \sum_{k=1}^K \mathbf{H}_k \mathbf{F}_k \mathbf{s}_k + \mathbf{w} \in \mathbb{C}^{N_R},$$

where $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_k}$ denotes the uplink channel between the k th user and the AP, $\mathbf{F}_k \in \mathbb{C}^{N_k \times d_k}$ denotes the precoding matrix for the k th user, d_k denotes the number of spatial streams user k is sending to the AP (and $d_k \leq N_k$), $\mathbf{s}_k \in \mathbb{C}^{d_k}$ denotes the transmitted signal for the k th user, $\mathbf{w} \in \mathbb{C}^{N_R}$ denotes the AWGN, with N_k denoting the number of antennas at the k th user and N_R denotes the number of antennas at the AP. It is assumed that $E[\mathbf{s}_k \mathbf{s}_k^H] = \mathbf{I}_{N_k}$. Let

$$x_k = \mathbf{F}_k \mathbf{s}_k,$$

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so

$$\begin{aligned}
E[\|\mathbf{x}_k\|_2^2] &= \mathbf{F}_k E[\mathbf{s}_k \mathbf{s}_k^H] \mathbf{F}_k^H \\
&= \mathbf{F}_k \mathbf{F}_k^H \\
&= \|\mathbf{F}_k\|_F^2 \leq P_k,
\end{aligned}$$

where P_k is the transmit power from the k th user. If SIC is used at the receiver, then the achievable rate at the receiver is

$$R_{sum} = \sum_{k=1}^K \log_2 \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_k \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{j \neq k} \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H \right)^{-1} \mathbf{F}_k^H \mathbf{H}_k^H \right)$$

Then it seems we can formulate the latency minimization problem as

$$\begin{aligned}
&\min_{\mathbf{F}_k, B_k, \forall k} \sum_{k=1}^K \frac{B_k}{R_{sum}} \\
&s.t. \quad \|\mathbf{F}_k\|_F^2 \leq P_k, \forall k, \\
&\quad B_k \in \{B_{k,\min}, \dots, B_{k,\max}\}.
\end{aligned} \tag{1}$$

Possible extension include optimizing not with the sum rate, but using rate such as

1. Max-min rate:

$$\max_{\mathbf{F}_k, \forall k} \min_k R_k.$$

This has a strong fairness guarantee, often leads to equal rates across all users if we can find a feasible solution. This is the worst-case that Carrson was talking about.

2. The proportional fairness:

$$\max_{\mathbf{F}_k, \forall k} \sum_{k=1}^K \log(R_k),$$

which balances the throughput and fairness.

3. Weighted fairness

$$\max_{\mathbf{F}_k, w_k, \forall k} \sum_{k=1}^K w_k \log(R_k),$$

where w_k are weights for QoS differentiation, i.e., higher w_k is, more resource will be allocated the k th user.

B. Uplink Latency Minimization Using RSMA

III. SIMULATION RESULTS

IV. CONCLUSION

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